

Title:

Smart Medical Lab Test Booking, Report & Follow-up Management System

Design and Planning:

1. Technical Stack

Backend:

- Java
- Spring Boot
- REST APIs

Database:

- MySQL

Frontend:

- HTML, CSS, JavaScript

Tools:

- Git & GitHub
- Postman (API testing)
- IDE (VS Code / IntelliJ IDEA)

2. Project Modules

Patient Module

- Registration and Login
- Search and Book Lab Tests
- Select Collection Method and Time Slot
- Track Sample Status
- View and Manage Reports
- Compare Test Results
- Follow-up Reminder and Next Test Booking

Laboratory Admin Module

- Manage Laboratory Details
- Manage Tests and Prices
- Manage Slots and Bookings
- Update Sample Status
- Upload Test Reports

- Manage Patient Test Records

Report Management Module

- Digital Report Storage
- Report Comparison
- Simple Result Trend Display
- Controlled Report Sharing

3. Functional Requirements

- User registration and authentication.
- Patients can search and book lab tests.
- Patients can select home collection or lab visit.
- System manages available slots and bookings.
- Lab staff can update sample and report status.
- Patients can view and download reports.
- System supports result comparison and follow-up reminders.

4. Non-Functional Requirements

- Security: Protect user and report data through authentication and access control.
- Performance: Provide quick response for booking and report operations.
- Reliability: Maintain accurate booking and report information.
- Usability: Provide a simple interface for patients and lab staff.
- Scalability: Allow adding more labs, tests, and users in future.